

Qatar-1b

R-2742

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210924 · created 2026-09-02T01:28:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459481.6971 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.029
Transit depth (Rp/Rs)^2	2.41 +/- 0.9 [%]
Orbital Inclination (inc)	88.78 +/- 1.31
Airmass coefficient 1 (a1)	0.978 +/- 0.022
Airmass coefficient 2 (a2)	0.014 +/- 0.052
Scatter in the residuals of the lightcurve fit is	2.23 %
Best Comparison Star	#1 - [414, 118]
Optimal Method	PSF photometry
Transit Duration (day)	0.0828 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.155 ± 0.029 vs 0.1463 (deviation 0.25σ)
Duration fit vs published	0.0828 d vs 0.0692 d (deviation 4.08σ)
Mid-time O-C	-2.8 min
Depth SNR	4.5
Residual scatter	2.23 %

Light curve

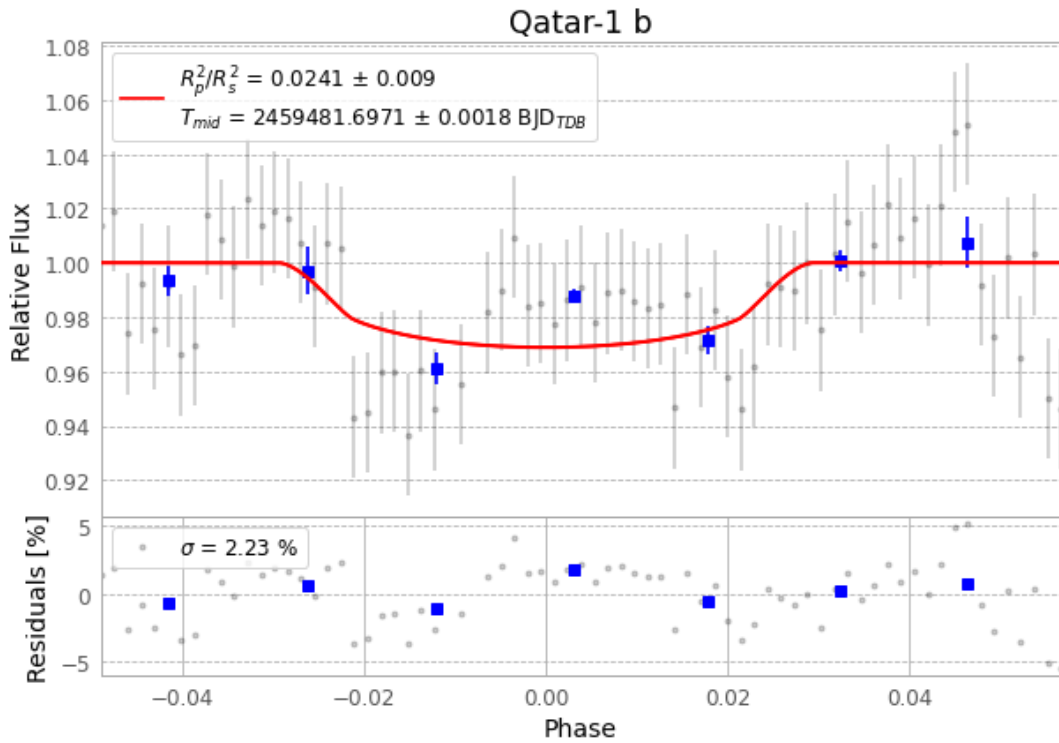


Figure 1 — FinalLightCurve_Qatar-1 b_2021-09-23.png (SHA-256 f62258856cd7c952...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [434, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 58019d72b892a6b1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1210924030012.FITS → Qatar-1210924063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210924.log (d8811f2f67f2...), FinalLightCurve_Qatar-1 b_2021-09-23.png (f62258856cd7...), FinalLightCurve_Qatar-1 b_2021-09-23.csv (be889dc02532...)

Compute resources

Wall time 136.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:28Z.